

# Category Theory and Functional Programming

## Functors

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# Preliminaries

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## Convention

The number assigned to chapters, examples, exercises, figures, pages, sections, and theorems on these slides correspond to the numbers assigned in the textbook (Abramsky and Tzevelekos 2011).

# Outline

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Introduction

Definition of a Functor

Examples of Functors

Functors in Haskell

Binary Functors

Small, Large and Locally Small Categories

The Category of Small Categories

Contravariance

Hom-Functors

Properties of Functors

References

# Introduction

# Introduction

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## Question

What about of morphisms between categories?

# Introduction

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## Question

What about of morphisms between categories?

*Answer:* Of course, them are functors.

## Definition of a Functor

# Definition of a Functor

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## Definition

A **(covariant) functor**  $F : \mathcal{C} \rightarrow \mathcal{D}$  between categories  $\mathcal{C}$  and  $\mathcal{D}$  is a mapping of objects to objects and arrows to arrows, that is,<sup>†</sup>

$$\begin{aligned} F_0 : \text{Obj}(\mathcal{C}) &\rightarrow \text{Obj}(\mathcal{D}) && \text{(object-map),} \\ F_1 : \text{Ar}(\mathcal{C}) &\rightarrow \text{Ar}(\mathcal{D}) && \text{(arrow-map),} \end{aligned}$$

which for all objects  $A$ ,  $B$  and  $C$  in  $\text{Obj}(\mathcal{C})$  and for all arrows  $A \xrightarrow{f} B$  and  $B \xrightarrow{g} C$  in  $\text{Ar}(\mathcal{C})$ , satisfies the **functoriality conditions**

$$\begin{aligned} F_1(g \circ f) &= (F_1 g) \circ (F_1 f) && \text{(preservation of compositions),} \\ F_1 \text{id}_A &= \text{id}_{(F_0 A)} && \text{(preservation of identities).} \end{aligned}$$

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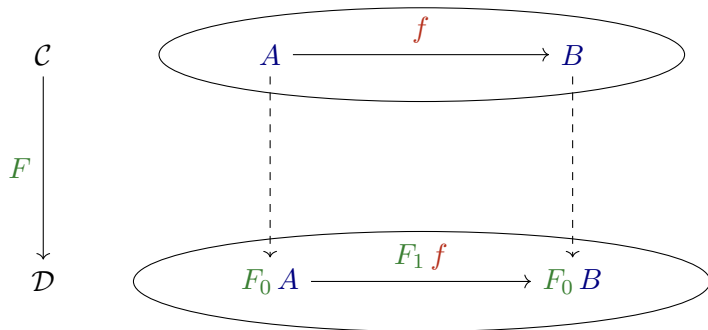
<sup>†</sup>The textbook does not use  $F_0$  and  $F_1$  but  $F$ .



# Definition of a Functor

## Observation

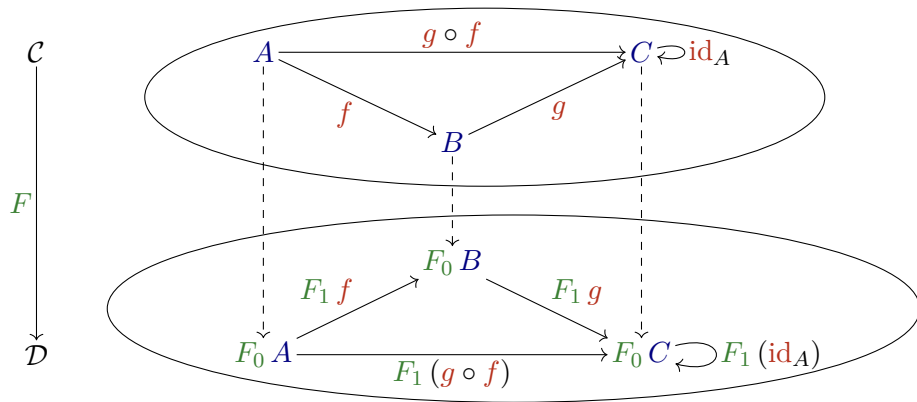
The functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  maps objects and arrows of  $\mathcal{C}$  to objects and arrows of  $\mathcal{D}$ , respectively.



# Definition of a Functor

## Observation

The functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  preserves domains and codomains, identity arrows, and composition. It also maps each commutative diagram in  $\mathcal{C}$  into a commutative diagram in  $\mathcal{D}$ .



# Definition of a Functor

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## Observation

Given a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$ , that is,

$$F_0 : \text{Obj}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{D}),$$

$$F_1 : \text{Ar}(\mathcal{C}) \rightarrow \text{Ar}(\mathcal{D}),$$

for all  $A, B$  in  $\text{Obj}(\mathcal{C})$ , there is the map

$$F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(F_0 A, F_0 B),$$

and for all  $f : A \rightarrow B$ ,

$$F_{A,B} f : F_0 A \rightarrow F_0 B.$$

# Examples of Functors

# Examples of Functors

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## Example

Let  $\mathcal{P}S$  be the power set of the set  $S$ . The **(covariant) power set functor**

$P : \mathbf{Set} \rightarrow \mathbf{Set}$ , is defined by

# Examples of Functors

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## Example

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$$P_0 X := \mathcal{P} X$$

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$$P_0 X := \mathcal{P} X$$

$$P_1 : \mathbf{Ar}(\mathbf{Set}) \rightarrow \mathbf{Ar}(\mathbf{Set})$$

$$P_{X,Y} : \mathbf{Set}(X, Y) \rightarrow \mathbf{Set}(P_0 X, P_0 Y)$$

$$P_{X,Y} f : \mathcal{P} X \rightarrow \mathcal{P} Y$$

$$P_{X,Y} f S := f(S) = \{ f(x) \mid x \in S \}$$

(continued on next slide)

# Examples of Functors

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## Example (continuation)

Let  $X = \{0, 1\}$ ,  $Y = \{\emptyset, X\}$  and  $f : X \rightarrow Y$  defined by  $f(0) = \emptyset$  and  $f(1) = X$ . Then,

$$P_0 : \text{Obj}(\mathbf{Set}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$P_0 X := \mathcal{P} X = \{\emptyset, \{0\}, \{1\}, X\},$$

$$P_{X,Y} f : \mathcal{P} X \rightarrow \mathcal{P} Y$$

$$P_{X,Y} f \emptyset := f(\emptyset) = \emptyset,$$

$$P_{X,Y} f \{0\} := f(\{0\}) = \{\emptyset\},$$

$$P_{X,Y} f \{1\} := f(\{1\}) = \{X\},$$

$$P_{X,Y} f \{0, 1\} := f(\{0, 1\}) = \{\emptyset, X\}.$$



# Examples of Functors

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## Example

Let  $(P, \preceq)$  and  $(Q, \preceq)$  be two pre-orders seen as categories, denoted  $\mathcal{P}$  and  $\mathcal{Q}$ , respectively. A functor  $F : \mathcal{P} \rightarrow \mathcal{Q}$  is defined by

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$$F_0 : \text{Obj}(\mathcal{P}) \rightarrow \text{Obj}(\mathcal{Q})$$

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$$F_{A,B} : \mathcal{P}(A, B) \rightarrow \mathcal{Q}(F_0 A, F_0 B)$$

$$F_{A,B} f : F_0 A \rightarrow F_0 B$$

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$$F_{A,B} f : F_0 A \rightarrow F_0 B$$

Since  $\mathcal{P}(A, B)$  and  $\mathcal{Q}(F_0 A, F_0 B)$  have at most an arrow, the map  $F_{A,B}$  exists iff

$$A \preceq B \quad \text{implies} \quad F_0 A \preceq F_0 B.$$

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That is, a functor  $F : \mathcal{P} \rightarrow \mathcal{Q}$  is just a monotone map which sends, if exists, the unique arrow  $A \rightarrow B$  to the unique arrow  $F_0 A \rightarrow F_0 B$ .

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Example from (Fong, Milewski and Spivak 2020, § 3.2.2).

# Examples of Functors

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## Example

Let  $(M, \cdot, \epsilon)$  and  $(N, \diamond, \mu)$  be two monoids seen as categories, denoted  $\mathcal{M}$  and  $\mathcal{N}$ , respectively. Let  $*$  be the only object in both categories. A functor  $F : \mathcal{M} \rightarrow \mathcal{N}$  is defined by

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$$F_0 : \text{Obj}(\mathcal{M}) \rightarrow \text{Obj}(\mathcal{N})$$

$$F_0 : \{*\} \rightarrow \{*\}$$

$$F_0 * = *$$



# Examples of Functors

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$$F_1 : \text{Ar}(\mathcal{M}) \rightarrow \text{Ar}(\mathcal{N})$$

$$F_{*,*} : \mathcal{P}(*, *) \rightarrow \mathcal{Q}(F_0 *, F_0 *)$$

$$F_{*,*} f : * \rightarrow *$$

# Examples of Functors

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$$F_{*,*} : \mathcal{P}(*, *) \rightarrow \mathcal{Q}(F_0 *, F_0 *)$$

$$F_{*,*} f : * \rightarrow *$$

The functor  $F$  must satisfy:

$$F_{*,*} (m_1 \cdot m_2) = (F_{*,*} m_1) \diamond (F_{*,*} m_2), \quad \text{for all } m_1, m_2 \text{ in } \mathcal{M},$$

$$F_{*,*} \epsilon = \mu.$$

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$$F_{*,*} \epsilon = \mu.$$

That is, a functor  $F : \mathcal{M} \rightarrow \mathcal{N}$  is just a monoid homomorphism.

# Examples of Functors

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## Example

The **identity functor**  $\text{Id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$  in a category  $\mathcal{C}$  is the functor that maps each object and each arrow of  $\mathcal{C}$  to itself.

# Examples of Functors

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## Example

Let  $F : \mathbf{Mon} \rightarrow \mathbf{Set}$  be the **forgetful functor** which

- (i) sends a monoid to its set of elements and
- (ii) sends a homomorphism between monoids to the corresponding function between sets.

# Examples of Functors

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## Example

Let  $[S]$  be the set of all finite lists of elements of  $S$ . The **list functor**

$\text{List} : \mathbf{Set} \rightarrow \mathbf{Set}$ , is defined by

# Examples of Functors

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## Example

Let  $[S]$  be the set of all finite lists of elements of  $S$ . The **list functor**

$\text{List} : \mathbf{Set} \rightarrow \mathbf{Set}$ , is defined by

$$\text{List}_0 : \text{Obj}(\mathbf{Set}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$\text{List}_0 X := [X]$$

# Examples of Functors

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$$\text{List}_1 : \text{Ar}(\mathbf{Set}) \rightarrow \text{Ar}(\mathbf{Set})$$

$$\text{List}_0 X := [X]$$

$$\text{List}_{X,Y} : \mathbf{Set}(X, Y) \rightarrow \mathbf{Set}(\text{List}_0 X, \text{List}_0 Y)$$

$$\text{List}_{X,Y} f : [X] \rightarrow [Y]$$

$$\text{List}_{X,Y} f [x_1, x_2, \dots, x_n] := [f(x_1), f(x_2), \dots, f(x_n)]$$



# Examples of Functors

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## Example

The **free monoid functor**  $\mathbf{MList} : \mathbf{Set} \rightarrow \mathbf{Mon}$  maps every set  $X$  to the free monoid over  $X$ .

# Examples of Functors

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Let  $(-) * (-)$  be the list concatenation function and let  $\varepsilon$  be the empty list, the functor is defined by

# Examples of Functors

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## Example

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Let  $(-)*(-)$  be the list concatenation function and let  $\varepsilon$  be the empty list, the functor is defined by

$$\mathbf{MList}_0 : \mathbf{Obj}(\mathbf{Set}) \rightarrow \mathbf{Obj}(\mathbf{Mon})$$

$$\mathbf{MList}_0 X := (\mathbf{List}_0 X, *, \varepsilon) = ([X], *, \varepsilon)$$

$$\mathbf{MList}_1 : \mathbf{Ar}(\mathbf{Set}) \rightarrow \mathbf{Ar}(\mathbf{Mon})$$

$$\mathbf{MList}_{X,Y} : \mathbf{Set}(X, Y) \rightarrow \mathbf{Mon}(\mathbf{MList}_0 X, \mathbf{MList}_0 Y)$$

$$\mathbf{MList}_{X,Y} f : ([X], *, \varepsilon) \rightarrow ([Y], *, \varepsilon)$$

$$\mathbf{MList}_{X,Y} f [x_1, x_2, \dots, x_n] := \mathbf{List}_{X,Y} f [x_1, x_2, \dots, x_n]$$

# Exercises

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## Exercise

Verify that functors  $F : \mathbf{2}_{\Rightarrow} \rightarrow \mathbf{Set}$  correspond to directed graphs (textbook, Exercise 45).

# Functors in Haskell

# Functors in Haskell

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Introduction via Maybe  
(Whiteboard).

# Functors in Haskell

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Introduction via Maybe  
(Whiteboard).

The typeclass Functor

```
class Functor f where  
    fmap :: (a -> b) -> f a -> f b
```

# Functors in Haskell

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## Example

The polymorphic type constructor `Maybe` is a functor whose instance is defined by

```
instance Functor Maybe where
    fmap _ Nothing  = Nothing
    fmap f (Just a) = Just (f a)
```



# Functors in Haskell

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## Example

The polymorphic type constructor `Maybe` is a functor whose instance is defined by

```
instance Functor Maybe where
  fmap _ Nothing  = Nothing
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```

## Exercise

Show that the `Maybe` functor satisfies the functoriality conditions.

# Functors in Haskell

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## Example

`ReadInt` is a type constructor that turns any type `a` into a new type that reads a value of `Int` to create a value of `a` (Fong, Milewski and Spivak 2020, Example 3.41).

```
data ReadInt a = MkReadInt (Int -> a)
```

# Functors in Haskell

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## Example

`ReadInt` is a type constructor that turns any type `a` into a new type that reads a value of `Int` to create a value of `a` (Fong, Milewski and Spivak 2020, Example 3.41).

```
data ReadInt a = MkReadInt (Int -> a)
```

`ReadInt` is a functor via the following instance.

```
instance Functor ReadInt where
  fmap f (MkReadInt g) = MkReadInt (f . g)
```

# Functors in Haskell

---

## Example

The (binary) function type  $(\rightarrow) :: a \rightarrow b \rightarrow (a \rightarrow b)$  is a functor.

```
instance Functor ((->) a) where
    fmap f g = f . g
```

Note that  $\text{fmap} :: (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow (a \rightarrow c)$ .

# Functors in Haskell

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## Exercise

To define an instance of `Functor` for the (binary) product type `(,) :: a -> b -> (a,b)`.

# Functors in Haskell

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## Example

Recall that terminal object (unit type) in [Haskell](#) is `() :: ()`. We can define a constant functor by

```
data CUnit a = MkCU ()

instance Functor CUnit where
    fmap f (MkCU ()) = MkCU ()
```

# Functors in Haskell

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## Exercise

Given a constant “functor” defined by

```
data CBool a = MkCB Bool

instance Functor CBool where
    fmap f (MkCB True)  = MkCB False
    fmap f (MkCB False) = MkCB True
```

Is CBool really a functor?

# Functors in Haskell

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## Exercise

We define a constant functor by

```
data CInt a = MkCI Int
```

Show that the polymorphic type constructor `CInt` can be given the structure of a functor by saying how it lifts morphisms. That is, provide a [Haskell](#) function `mapCInt` of the type  $(a \rightarrow b) \rightarrow (CInt\ a \rightarrow CInt\ b)$  (Fong, Milewski and Spivak 2020, Exercise 3.46).



# Functors in Haskell

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## Exercise

For each of the following type constructors, define two versions of `fmap`, one of which has a corresponding functor `Hask`  $\rightarrow$  `Hask`, and one of which does not (Fong, Milewski and Spivak 2020, Exercise 3.48).

- (i) `data WithString a = WithStr (a, String)`
- (ii) `data ConstStr a = ConstStr String`
- (iii) `data List a = Nil | Cons (a, List a)`

# Binary Functors

# The Product Category

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## Definition

Let  $\mathcal{C}$  and  $\mathcal{D}$  be two categories. The **product category**  $\mathcal{C} \times \mathcal{D}$  is defined by:

- (i) Objects:  $(C, D)$ , where  $C$  and  $D$  are objects in  $\mathcal{C}$  and  $\mathcal{D}$ , respectively.
- (ii) Arrows:  $(C, D) \xrightarrow{(f, g)} (C', D')$ , where  $C \xrightarrow{f} C'$  and  $D \xrightarrow{g} D'$  are arrows in  $\mathcal{C}$  and  $\mathcal{D}$ , respectively.
- (iii) Composition

$$(f', g') \circ (f, g) := (f' \circ f, g' \circ g).$$

- (iv) Identities

$$\text{id}_{(C, D)} := (\text{id}_C, \text{id}_D).$$

# Definition of a Binary Functor

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## Definition

Let  $\mathcal{C}$ ,  $\mathcal{D}$  and  $\mathcal{E}$  be three categories. A **binary functor** (or **bifunctor**) is a functor whose domain is a product category, that is, a binary functor from  $\mathcal{C} \times \mathcal{D}$  to  $\mathcal{E}$  is a functor

$$F : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}.$$

## Example of Binary Functors

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### Example

The projection functors  $\mathcal{C} \xleftarrow{\pi_1} \mathcal{C} \times \mathcal{D} \xrightarrow{\pi_2} \mathcal{D}$  are binary functors.

# Example of Binary Functors

---

## Example

The projection functors  $\mathcal{C} \xleftarrow{\pi_1} \mathcal{C} \times \mathcal{D} \xrightarrow{\pi_2} \mathcal{D}$  are binary functors.

(i) For  $\pi_1 : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C}$  we have:

$$(\pi_1)_0 : \text{Obj}(\mathcal{C} \times \mathcal{D}) \rightarrow \text{Obj}(\mathcal{C})$$

$$(\pi_1)_0 (C, D) := C$$

$$(\pi_1)_1 : \text{Ar}(\mathcal{C} \times \mathcal{D}) \rightarrow \text{Ar}(\mathcal{C})$$

$$(\pi_1)_{(C,D),(C',D')} : \text{Mor}_{\mathcal{C} \times \mathcal{D}}((C, D), (C', D')) \rightarrow \text{Mor}_{\mathcal{C}}((\pi_1)_0 (C, D), (\pi_1)_0 (C', D')),$$

$$(\pi_1)_{(C,D),(C',D')} (f, g) : C \rightarrow C'$$

$$(\pi_1)_{(C,D),(C',D')} (f, g) := f$$

## Example of Binary Functors

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### Example

The projection functors  $\mathcal{C} \xleftarrow{\pi_1} \mathcal{C} \times \mathcal{D} \xrightarrow{\pi_2} \mathcal{D}$  are binary functors.

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$$(\pi_1)_{(C,D),(C',D')} : \text{Mor}_{\mathcal{C} \times \mathcal{D}}((C, D), (C', D')) \rightarrow \text{Mor}_{\mathcal{C}}((\pi_1)_0(C, D), (\pi_1)_0(C', D')),$$

$$(\pi_1)_{(C,D),(C',D')}(\textcolor{red}{f}, \textcolor{red}{g}) : C \rightarrow C'$$

$$(\pi_1)_{(C,D),(C',D')}(\textcolor{red}{f}, \textcolor{red}{g}) := \textcolor{red}{f}$$

(ii) Similarly for  $\pi_2 : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{D}$ .

# The Product Functor

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## Definition

Let  $\mathcal{C}$  be a category with binaries products, and let  $\mathcal{C} \times \mathcal{C}$  be the product category of  $\mathcal{C}$  with itself. The **product functor**  $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  is a binary functor defined by



# The Product Functor

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## Definition

Let  $\mathcal{C}$  be a category with binary products, and let  $\mathcal{C} \times \mathcal{C}$  be the product category of  $\mathcal{C}$  with itself. The **product functor**  $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  is a binary functor defined by

$$\times_0 : \text{Obj}(\mathcal{C} \times \mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$$

$$\times_0(A, B) := A \times B \text{ (binary product)}$$

$$\times_1 : \text{Ar}(\mathcal{C} \times \mathcal{C}) \rightarrow \text{Ar}(\mathcal{C})$$

$$\times_{(A, A'), (B, B')} : \mathcal{C} \times \mathcal{C}((A, A'), (B, B')) \rightarrow \mathcal{C}(\times_0(A, A'), \times_0(B, B'))$$

$$\times_1(f : A \rightarrow B, g : A' \rightarrow B') : A \times A' \rightarrow B \times B'$$

$$\times_1(f : A \rightarrow B, g : A' \rightarrow B') := f \times g \text{ (product morphism)}$$

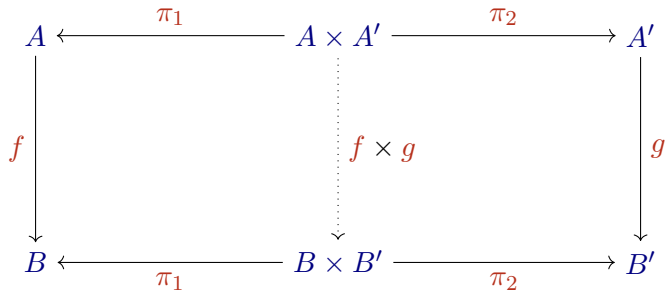
where  $f \times g := \langle f \circ \pi_1, g \circ \pi_2 \rangle$ .

(continued on next slide)

# The Product Functor

## Definition (continuation)

That is, both squares in the following diagram commute.



$$\begin{pmatrix} f \circ \pi_1 = \pi_1 \circ (f \times g) \\ g \circ \pi_2 = \pi_2 \circ (f \times g) \end{pmatrix}$$

# $N$ -Ary Functors

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## Observation

Binary functors can be generalised to  $n$ -ary functors.

# Small, Large and Locally Small Categories

# Small and Large Categories

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## Introduction

Before defining a category of categories, we need to classify the categories in small and large for avoiding that it be an object of itself.

# Small and Large Categories

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## Definition

A category is **small** iff **both** the collection of its objects and the collection of its arrows are **sets**. Otherwise, the category is **large** (Awodey 2010).

# Small and Large Categories

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## Example

The finite categories  $\mathbf{1}, \mathbf{2}, \dots, \mathbf{n}$ , a monoid viewed as a category, and a pre-order viewed as a category are small categories.

# Small and Large Categories

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## Example

The finite categories  $\mathbf{1}, \mathbf{2}, \dots, \mathbf{n}$ , a monoid viewed as a category, and a pre-order viewed as a category are small categories.

## Example

The categories **Set**, **Pos**, **Mon**, **Grp** and **Top** are large categories.



# Locally Small Categories

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## Definition

A category  $\mathcal{C}$  is **locally small** iff for all objects  $A, B$  the collection  $\mathcal{C}(A, B)$  is a **set** (Awodey 2010).

# Locally Small Categories

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## Definition

A category  $\mathcal{C}$  is **locally small** iff for all objects  $A, B$  the collection  $\mathcal{C}(A, B)$  is a **set** (Awodey 2010).

## Observation

- Recall from the previous conventions that if the collection  $\mathcal{C}(A, B)$  is a set it is called a hom-set and it is denoted  $\text{hom}_{\mathcal{C}}(A, B)$ .
- Also recall that in the textbook all the collections  $\mathcal{C}(A, B)$  are hom-sets.

# Locally Small Categories

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## Example

Any small category is locally small.

# Locally Small Categories

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## Example

Any small category is locally small.

## Example

The categories **Set**, **Pos**, **Mon**, **Grp** and **Top** are locally small categories.

# The Category of Small Categories

# The Category of Small Categories

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## Definition

The category  $\mathbf{Cat}$  is the category of small categories:

- (i) Objects: Small categories
- (ii) Arrows: Functors

(continued on next slide)

# The Category of Small Categories

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## Definition (continuation)

### (iii) Composition of functors

Let  $\mathcal{C}$ ,  $\mathcal{D}$  and  $\mathcal{E}$  be small categories and let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{E}$  be two functors, then

$$G \circ F : \mathcal{C} \rightarrow \mathcal{E},$$

$$(G \circ F)_0 : \text{Obj}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{E})$$

$$(G \circ F)_0 A := G_0 (F_0 A),$$

$$(G \circ F)_1 : \text{Ar}(\mathcal{C}) \rightarrow \text{Ar}(\mathcal{E}),$$

$$(G \circ F)_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{E}((G \circ F)_0 A, (G \circ F)_0 B)$$

$$(G \circ F)_{A,B} f : G_0 (F_0 A) \rightarrow G_0 (F_0 B)$$

$$(G \circ F)_{A,B} f := G_1 (F_1 f).$$

(continued on next slide)

# The Category of Small Categories

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## Definition (continuation)

### (iii) Composition of functors

That is,

$$G \circ F : \mathcal{C} \rightarrow \mathcal{E} := \begin{cases} A \mapsto G(F A), \\ f \mapsto G(F f). \end{cases}$$



# The Category of Small Categories

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## Definition (continuation)

### (iii) Composition of functors

That is,

$$G \circ F : \mathcal{C} \rightarrow \mathcal{E} := \begin{cases} A \mapsto G(F A), \\ f \mapsto G(F f). \end{cases}$$

### (iv) Identity functors

Let  $\mathcal{C}$  be a small category, then

$$\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C} := \begin{cases} A \mapsto A, \\ f \mapsto f. \end{cases}$$

# The Category of Small Categories

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## Observation

The category **Cat** is large and therefore it is not object of itself.

# Contravariance

# Introduction

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## Description

A **covariant** functor  $F$  preserves the direction of arrows, that is,

$$F_1 (f : A \rightarrow B) : F_0 A \rightarrow F_0 B.$$

A **contravariant** functor  $G$  reverses the direction of arrows, that is,

$$G_1 (f : A \rightarrow B) : G_0 B \rightarrow G_0 A.$$

# Contravariant Functors

## Definition

Let  $\mathcal{C}$  and  $\mathcal{D}$  be two categories. A **contravariant functor**  $G$  from  $\mathcal{C}$  to  $\mathcal{D}$  is a functor

$$G : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D} \quad (\text{or } \mathcal{C} \rightarrow \mathcal{D}^{\text{op}}),$$

$$G_0 : \text{Obj}(\mathcal{C}^{\text{op}}) \rightarrow \text{Obj}(\mathcal{D}) \quad (\text{object-map}),$$

$$G_1 : \text{Ar}(\mathcal{C}^{\text{op}}) \rightarrow \text{Ar}(\mathcal{D}) \quad (\text{arrow-map}),$$

$$G_{A,B} : \mathcal{C}^{\text{op}}(A, B) \rightarrow \mathcal{D}(G_0 B, G_0 A)$$

$$G_{A,B} f : G_0 B \rightarrow G_0 A,$$

which for all objects  $A, B$  and  $C$  in  $\text{Obj}(\mathcal{C}^{\text{op}})$  and for all arrows  $A \xrightarrow{f} B$  and  $B \xrightarrow{g} C$  in  $\text{Ar}(\mathcal{C}^{\text{op}})$ , satisfies the **functoriality conditions**

$$G_1(g \circ f) = (G_1 f) \circ (G_1 g) \quad (\text{preservation of compositions}),$$

$$G_1(\text{id}_A) = \text{id}_{(G_0 A)} \quad (\text{preservation of identities}).$$

# Contravariant Functors

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## Example

Let  $\mathcal{P}S$  be the power set of the set  $S$ . The **contravariant power set functor**

$\mathcal{P}^{\text{op}} : \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ , is defined by

# Contravariant Functors

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## Example

Let  $\mathcal{P} S$  be the power set of the set  $S$ . The **contravariant power set functor**

$P^{\text{op}} : \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ , is defined by

$$P_0^{\text{op}} : \text{Obj}(\mathbf{Set}^{\text{op}}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$P_0^{\text{op}} X := \mathcal{P} X$$

# Contravariant Functors

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## Example

Let  $\mathcal{P}S$  be the power set of the set  $S$ . The **contravariant power set functor**

$P^{\text{op}} : \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ , is defined by

$$P_0^{\text{op}} : \text{Obj}(\mathbf{Set}^{\text{op}}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$P_0^{\text{op}} X := \mathcal{P} X$$

$$P_1^{\text{op}} : \text{Ar}(\mathbf{Set}^{\text{op}}) \rightarrow \text{Ar}(\mathbf{Set})$$

$$P_{X,Y}^{\text{op}} : \mathbf{Set}^{\text{op}}(X, Y) \rightarrow \mathbf{Set}(P_0^{\text{op}} Y, P_0^{\text{op}} X)$$

$$P_{X,Y}^{\text{op}} f : \mathcal{P} Y \rightarrow \mathcal{P} X$$

$$P_{X,Y}^{\text{op}} f T := f^{-1}(T) = \{ x \in X \mid f(x) \in T \}$$



# Hom-Functors

# Hom-Functors

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## Definition (first notation)

Let  $\mathcal{C}$  be a locally small category and let  $A$  be an object of  $\mathcal{C}$ . The **covariant Set-valued hom-functor**  $\mathcal{C}(A, -)$  is defined by

$$\mathcal{C}(A, -) : \mathcal{C} \rightarrow \mathbf{Set},$$

$$\mathcal{C}(A, -)_0 : \mathrm{Obj}(\mathcal{C}) \rightarrow \mathrm{Obj}(\mathbf{Set})$$

$$\mathcal{C}(A, C)_0 := \mathcal{C}(A, C),$$

$$\mathcal{C}(A, -)_1 : \mathrm{Ar}(\mathcal{C}) \rightarrow \mathrm{Ar}(\mathbf{Set})$$

$$\mathcal{C}(A, -)_{C,D} : \mathcal{C}(C, D) \rightarrow \mathbf{Set}(\mathcal{C}(A, -)_0 C, \mathcal{C}(A, -)_0 D)$$

$$\mathcal{C}(A, f)_{C,D} : \mathcal{C}(A, C) \rightarrow \mathcal{C}(A, D)$$

$$\mathcal{C}(A, f)_{C,D} g := f \circ g.$$

# Hom-Functors

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## Definition (first notation)

Let  $\mathcal{C}$  be a (locally small) category and let  $B$  be an object of  $\mathcal{C}$ . The **contravariant Set-valued hom-functor**  $\mathcal{C}(-, B)$  is defined by

$$\mathcal{C}(-, B) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set},$$

$$\mathcal{C}(-, B)_0 : \text{Obj}(\mathcal{C}^{\text{op}}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$\mathcal{C}(C, B)_0 := \mathcal{C}(C, B),$$

$$\mathcal{C}(-, B)_1 : \text{Ar}(\mathcal{C}^{\text{op}}) \rightarrow \text{Ar}(\mathbf{Set})$$

$$\mathcal{C}(-, B)_{C,D} : \mathcal{C}^{\text{op}}(C, D) \rightarrow \mathbf{Set}(\mathcal{C}(-, B)_0 D, \mathcal{C}(-, B)_0 C)$$

$$\mathcal{C}(f, B)_{C,D} : \mathcal{C}(D, B) \rightarrow \mathcal{C}(C, B)$$

$$\mathcal{C}(f, B)_{C,D} g := g \circ f.$$

## Exercise

Let  $\mathcal{C}$  be a (locally small) category. Spell out the definition of the set-valued hom-functor  $\mathcal{C}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$ . Verify carefully that it is a functor (textbook, Exercise 47).

# Hom-Functors

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## Notation

Recall that if  $\mathcal{C}$  is a locally small category the collection of arrows of an object  $A$  to an object  $B$  is a **set** and it is denoted by  $\text{hom}_{\mathcal{C}}(A, B)$ , that is,

$$\text{hom}_{\mathcal{C}}(A, B) := \left\{ f \in \text{Ar}(\mathcal{C}) \mid A \xrightarrow{f} B \right\} =: \mathcal{C}(A, B).$$

# Hom-Functors

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## Definition (second notation)

Let  $\mathcal{C}$  be a locally small category and let  $A$  be an object of  $\mathcal{C}$ . The **covariant Set-valued hom-functor**  $\text{hom}_{\mathcal{C}}(A, -)$  is defined by

$$\text{hom}_{\mathcal{C}}(A, -) : \mathcal{C} \rightarrow \mathbf{Set},$$

$$\text{hom}_{\mathcal{C}}(A, -)_0 : \text{Obj}(\mathcal{C}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$\text{hom}_{\mathcal{C}}(A, C)_0 := \text{hom}_{\mathcal{C}}(A, C)$$

$$\text{hom}_{\mathcal{C}}(A, -)_1 : \text{Ar}(\mathcal{C}) \rightarrow \text{Ar}(\mathbf{Set})$$

$$\text{hom}_{\mathcal{C}}(A, -)_{C,D} : \text{hom}_{\mathcal{C}}(C, D) \rightarrow \mathbf{Set}(\text{hom}_{\mathcal{C}}(A, C), \text{hom}_{\mathcal{C}}(A, D))$$

$$\text{hom}_{\mathcal{C}}(A, f : C \rightarrow D) : \text{hom}_{\mathcal{C}}(A, C) \rightarrow \text{hom}_{\mathcal{C}}(A, D)$$

$$\text{hom}_{\mathcal{C}}(A, f : C \rightarrow D) g := f \circ g$$

# Hom-Functors

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## Definition (second notation)

Let  $\mathcal{C}$  be a (locally small) category and let  $B$  be an object of  $\mathcal{C}$ . The **contravariant Set-valued hom-functor**  $\text{hom}_{\mathcal{C}}(-, B)$  is defined by

$$\text{hom}_{\mathcal{C}}(-, B) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set},$$

$$\text{hom}_{\mathcal{C}}(-, B)_0 : \text{Obj}(\mathcal{C}^{\text{op}}) \rightarrow \text{Obj}(\mathbf{Set})$$

$$\text{hom}_{\mathcal{C}}(C, B)_0 := \text{hom}_{\mathcal{C}}(C, B)$$

$$\text{hom}_{\mathcal{C}}(-, B)_1 : \text{Ar}(\mathcal{C}^{\text{op}}) \rightarrow \text{Ar}(\mathbf{Set})$$

$$\text{hom}_{\mathcal{C}}(-, B)_{C,D} : \text{hom}_{(\mathcal{C}^{\text{op}})}(C, D) \rightarrow \mathbf{Set}(\text{hom}_{\mathcal{C}}(D, B), \text{hom}_{\mathcal{C}}(C, B))$$

$$\text{hom}_{\mathcal{C}}(f : C \rightarrow D, B) : \text{hom}_{\mathcal{C}}(D, B) \rightarrow \text{hom}_{\mathcal{C}}(C, B)$$

$$\text{hom}_{\mathcal{C}}(f : C \rightarrow D, B) g := g \circ f$$

# Properties of Functors



# Faithful and Full Functors

---

## Definition

Let  $\mathcal{C}$  and  $\mathcal{D}$  be (locally small) categories and let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor.

- (i) The functor  $F$  is **faithful** iff each map  $F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(F_0 A, F_0 B)$  is injective.
- (ii) The functor  $F$  is **full** iff each map  $F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(F_0 A, F_0 B)$  is surjective.

# Faithful and Full Functors

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## Example

The forgetful functor  $F : \mathbf{Mon} \rightarrow \mathbf{Set}$  is faithful, but not full (explanation in the whiteboard).

# Faithful and Full Functors

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## Example

The forgetful functor  $F : \mathbf{Mon} \rightarrow \mathbf{Set}$  is faithful, but not full (explanation in the whiteboard).

Let  $(M, \cdot, 1_M)$  and  $(N, *, 1_N)$  be two monoids and let  $f : M \rightarrow N$  be a homomorphism between them.

- Since  $F_1 f = f$ , the map  $F_1$  is injective.
- If  $g : M \rightarrow N$  is any function in  $\mathbf{Set}$  such that  $g(1_M) \neq 1_N$ , then  $g$  is not a homomorphism between  $(M, \cdot, 1_M)$  and  $(N, *, 1_N)$ . Therefore the map  $F_1$  is not surjective.

# Faithful and Full Functors

---

## Exercise

Show that the free monoid functor  $\mathbf{MList} : \mathbf{Set} \rightarrow \mathbf{Mon}$  is faithful, but not full.

## Exercise (1.3.5.2)

Let  $\mathcal{C}$  be a category with binary products such that, for each pair of objects  $A, B$ ,

$$\mathcal{C}(A, B) \neq \emptyset. \quad (*)$$

- (i) Show that the product functor  $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  is faithful.
- (ii) Would  $-\times-$  still be faithful in the absence of condition  $(*)$ ?

# Preservation and Reflection

---

## Definition

Let  $P$  be a property of arrows and let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor.

- (i) The functor  $F$  **preserves** the property  $P$  iff  
if  $f$  satisfies  $P$  then  $F_1 f$  satisfies  $P$ .
- (ii) The functor  $F$  **reflects** the property  $P$  iff  
if  $F_1 f$  satisfies  $P$  then  $f$  satisfies  $P$ .

# Preservation and Reflection

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## Example

Show that all functors preserve isomorphisms.

# Preservation and Reflection

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## Example

Show that all functors preserve isomorphisms.

## Example

Show that full and faithful functors reflect isomorphisms.

## References



## References

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